

that can be read and written in by using a USB or similar connection. Since this is not the primary function of such devices, they are overlooked as potential threats. But the fact is, if an endpoint can easily read and execute data from the device, then it can really pose as much of a threat as a thumb drive. Such devices include MP3 players, digital cameras, printers, scanners, digital picture frames, etc.

Preventive measures

1. Implement and enforce different asset control policies that keep a check on what devices can enter the environment and, also, when they can enter.
2. There should be frequent reminders about all the policies that deal with these controls.

The Trojan human

Just like the Trojan horse, the Trojan human also enters a business in disguise. The Trojan human could be in business attire or dressed like a repairman (telecom, appliance, HVAC, etc). Such tricksters are known to have penetrated some of the most secure environments (including server rooms).

Preventive measures

1. Continuous reminders should be sent to the employees regarding the best practices to follow when authorising third parties.
2. Sources accessing our system must be properly identified and validated by asking questions, rather than making assumptions.

Optical media

Most people are surprised to learn that even optical media like CDs can be used to infect a network. CDs labelled with the name of popular recording artists are used as a guise. Once the CD is inserted into the system, the classified information can be accessed using authorised credentials, and the secure data present on the system can be easily stored on the CD in encrypted archives. So, recordable media that appears to be legitimate can also be used to piggyback data in and out of networks.

Preventive measures

1. Implement and enforce different asset control policies that keep a check on the devices that can enter the environment and when.
2. There should be frequent reminders about the policy and the kind of controls incorporated.

E-mail

E-mail, too, is often misused. Messages with confidential information can be forwarded to any external target. In addition, the e-mails themselves can let in nasty viruses. One e-mail could phish for the access credentials of an employee and later, these stolen credentials can be leveraged in a second-stage attack.

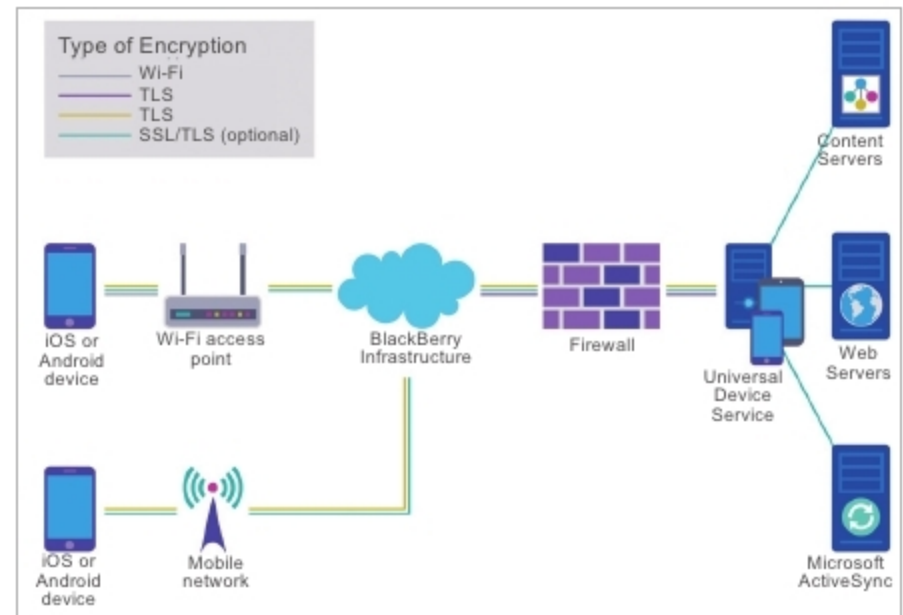


Figure 1: Different types of encryption implemented as part of network security
(Image source: *googleimages.com*)

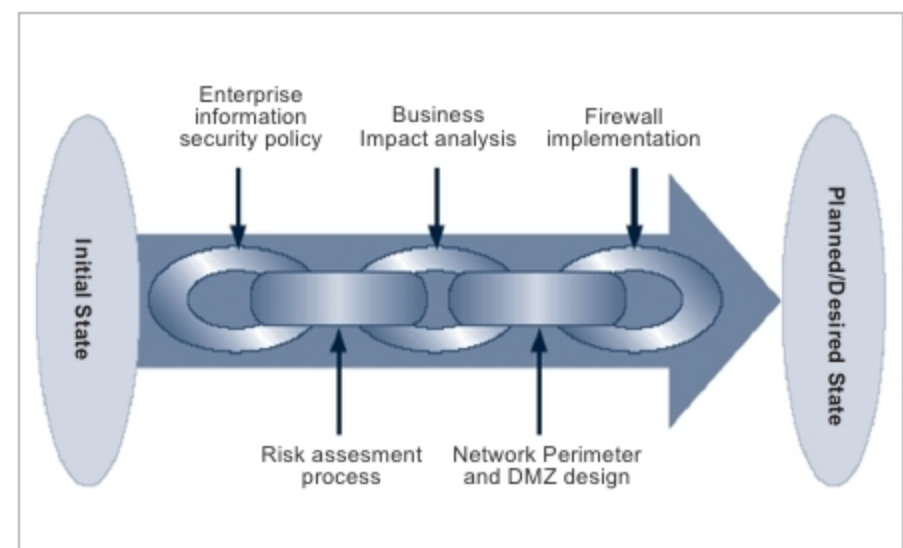


Figure 2: Information Security Program Chain (Image source: [googleimages.com](#))

Preventive measures

1. Always identify the sender of the mail by using technologies such as PGP. Even a simple set of questions before sending any sensitive information will work.
2. Access control should be enforced for broad, alias-based e-mail addresses.
3. Reminders about the security policies should be continuously sent out to employees.

Different open source tools to secure a network

Let's have a quick look at some of the important open source tools that help us secure a local network.

Nmap: This is a free and open source network scanner tool that is used to discover the various hosts and services present on a computer network by sending different packets and analysing the response for each of them. It offers a couple of important features that play a great role in protecting our network from external infections. It features the powerful NSE scripts that can help detect misconfiguration, vulnerabilities, and security related information about network services. Nmap can easily address network conditions like latency and congestion during a scan.